



Comparing the Beta Estimation Methods of Fixed-Time Periods, Rolling Windows, and Random Forests on Futures Contracts

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1 Abstract

This paper compares three unique methods of beta estimation for futures contracts: fixed-time period regressions, rolling window regressions, and random forests. By examining how these different approaches estimate beta, this research will identify which specific method is most optimal.

Each estimation approach which is analyzed in this study offers distinct advantages: fixed-time period regressions provide consistency, rolling windows incorporate temporal changes, and Random Forest models capture more non-linear relationships. However, fixed-time periods inability to dynamically change, leads me to hypothesize that rolling window regression and random forest models, will outperform the traditional model in its ability to calculate risk.

The findings of this paper can have significant implications for the AlgoGators Investment Fund and can change how the fund approaches the aspect of risk calculation, ultimately leading to a more secure and safe portfolio.

2 Introduction

Beta measures how much an asset moves with respect to the overall market and can be utilized to assess the risk of an investment. For futures contracts, accurate beta estimation is crucial due to their leveraged nature and prominent use within hedging strategies. Traditional beta estimation uses fixed time periods, which assumes that the

relationship between the asset and the market remains constant over time. However, this is often not the case, as changing market conditions, investor behavior, and economic factors can cause this relationship to shift. Because of this conflict, more and more funds are exploring more unique methods to estimate beta.

3 Methodology

For data collection, I utilized AlgoGators' Data-Ngin, a powerful data tool within the fund that provides daily financial values spanning several years. In the event of missing data, the yfinance API was utilized to fill the gaps.

The focus was placed on three key futures contracts: E-mini-S&P 500 (ES), Crude Oil (CL), and Gold (GC). For each, I extracted their daily adjusted back-adjusted closing prices to account for rollover effects. These back-adjusted prices provide accurate daily price values which were suitable for analysis.

As beta measures an asset's sensitivity to movements in the overall market, a reliable benchmark is essential for its estimation. To serve this role, I selected the S&P 500 Index, one of the most widely recognized representations of the broader equity market. This benchmark provides the necessary comparative framework for evaluating the systematic risk of the chosen futures contracts. Additionally, for the random forest model I utilized volatility indicators obtained from CBOE volatility index as an input.

The dataset spans from January 1, 2013, to January 1, 2023, providing several advantages for robust beta estimation. This timeframe encompasses diverse market regimes while offering data that is recent enough to be relevant.

Once all the data was collected and cleaned, I was able to begin the actual beta estimation. To truly compare the movements of the contracts, the log returns were calculated instead of simply relying on the closing prices. The log returns allowed for more accurate comparison as it set a standard between the instruments despite the differences in price. All the data was then organized into an excel spreadsheet to begin calculations.

3.1 Fixed-Time Period Regression

The fixed-time period beta was calculated using the following formula for beta as found in the Capital Asset Pricing Model (CAPM) framework:

$$Beta = \beta = \frac{Cov(R_{futures}, R_{market})}{Var(R_{market})}$$

$R_{futures}$ represents the daily log returns of the future contracts, calculated as $\ln(P_t/P_{t-1})$

where P_t is the closing price on day t . R_{market} represents the daily log returns of the benchmark, computed similarly. Cov represents covariance and Var represents variance.

Ultimately, within Excel using the regression Analysis Tool-Pak add on, I was able to determine the fixed-time period regression beta values for all three futures contracts.

3.2 Rolling Windows Regression

For the rolling windows regression approach, a series of beta values were calculated over a 60-day rolling window period. This allowed the beta to adapt to changing market conditions throughout the sample period. This theoretically should provide a more responsive measure of the futures' contracts sensitivity in comparison to its static fixed-time period counterpart. The specific formula used was the same as the aforementioned one:

$$Beta = \beta = \frac{Cov(R_{futures}, R_{market})}{Var(R_{market})}$$

The rolling window beta was computed via Python thru the Pandas library, which resulted in efficient time-series calculations. For each trading day t , the beta was then estimated using the 60 most recent days of log returns. This resulted in a series of beta values; however, the first 59 days were lacking estimates due to the 60-day window just beginning.

3.3 Random Forest

For this part, a random forest regression approach was employed to try and enhance the beta estimation. This ML method predicts beta values with the assistance of an additional indicator: the CBOE volatility index. This provided a more dynamic alternative than the rolling window regression model. Unlike the fixed time-period and rolling window methods, which directly compute beta using the CAPM formula, the random forest approach predicts beta as a function of selected market variables. The target variable was the rolling window beta, which served as a proxy for the true time-varying beta. The random forest

regression was implemented in Python using the RandomForestRegressor class from the Scikit-learn library. It utilized hyperparameters tuned for financial time series: 100 estimators (decision trees), a maximum depth of 10 nodes, and controlled randomness through fixed random state seeding. The dataset was split into training (80%) and testing (20%) sets to maintain the temporal distribution characteristics.

4 Results

To evaluate the effectiveness and stability of each beta estimation method, the results are interpreted within the framework of the Capital Asset Pricing Model (CAPM). According to CAPM, the expected return of an asset is linearly related to its beta with respect to the market portfolio, as seen in the following formula:

$$\text{Expected Return of Investment} = R_f + \beta_i(ER_m - R_f)$$

R_f is the risk-free rate, β_i is the beta, and $ER_m - R_f$ is the market risk premium.

This formula enables testing the predictive accuracy of each method's estimated beta. Each beta is incorporated into the CAPM formula, using a risk-free rate of 0.0000461514 (the average daily 3-month Treasury Bill rate from FRED, 2013–2023) and the historical S&P 500 log returns, to compute the expected return of the futures contract. The market risk premium was incorporated by subtracting the daily risk-free rate from the S&P 500 daily log returns, calculated from back-adjusted closing prices, to reflect the market's excess return. These expected returns are then compared to the actual average returns. The

method yielding the lowest prediction errors, measured by mean absolute error (MAE) and mean squared error (MSE), is then deemed to be most effective at capturing the futures contract's true market sensitivity.

For clarity, I will first dive into the results of each of the individual beta estimation methods, and then I will go into the CAPM comparative results.

4.1 Fixed-Time Period Regression Results

The following table shows the beta results calculated in Excel, via the regression model:

Futures Contract Name	Fixed Time Period Beta Value
ES	0.967846
GC	-0.01041
CL	0.645916

Figure 1: Table of Fixed-Time Period Regression Result

The E-mini S&P 500 futures (ES) had a beta of 0.967846, which is very close to 1. This makes sense as it directly tracks the S&P 500 stock index.

Gold futures (GC) displayed a slightly negative beta of -0.01041, consistent with its safe-haven status that typically moves inversely to equity markets during uncertainty. The weak magnitude indicates limited correlation with the S&P 500.

Crude Oil futures (CL) exhibited a beta of 0.645916, reflecting moderate sensitivity to equity markets due to shared economic drivers.

4.2 Rolling Window Regression Results

The following graphs showcase the rolling regression plotted onto a graph, all developed within python:

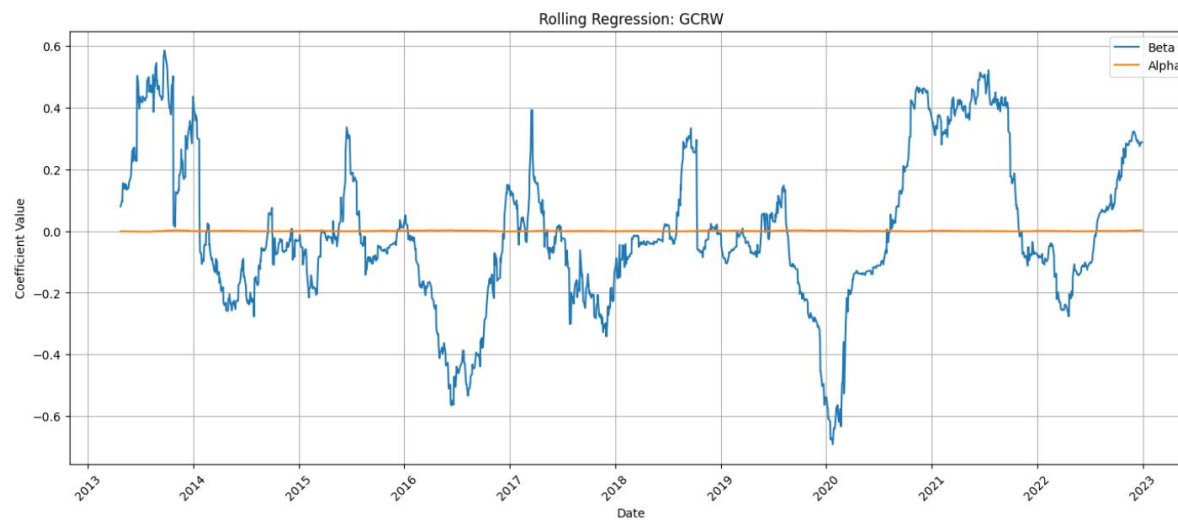


Figure 2: Rolling Regression of GC Futures Plot

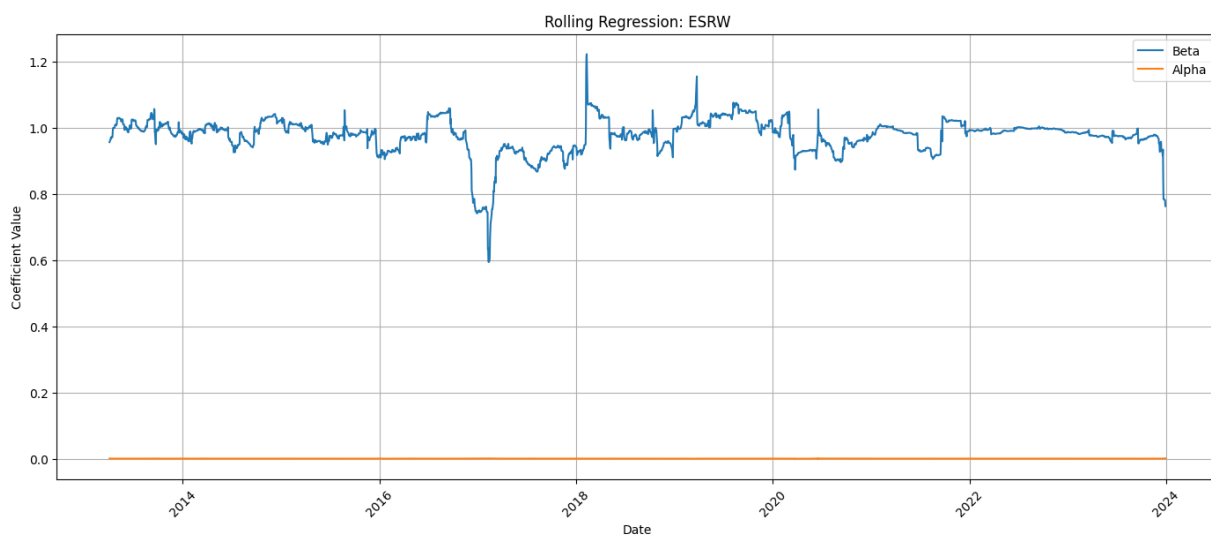


Figure 3: Rolling Regression of ES Futures Plot

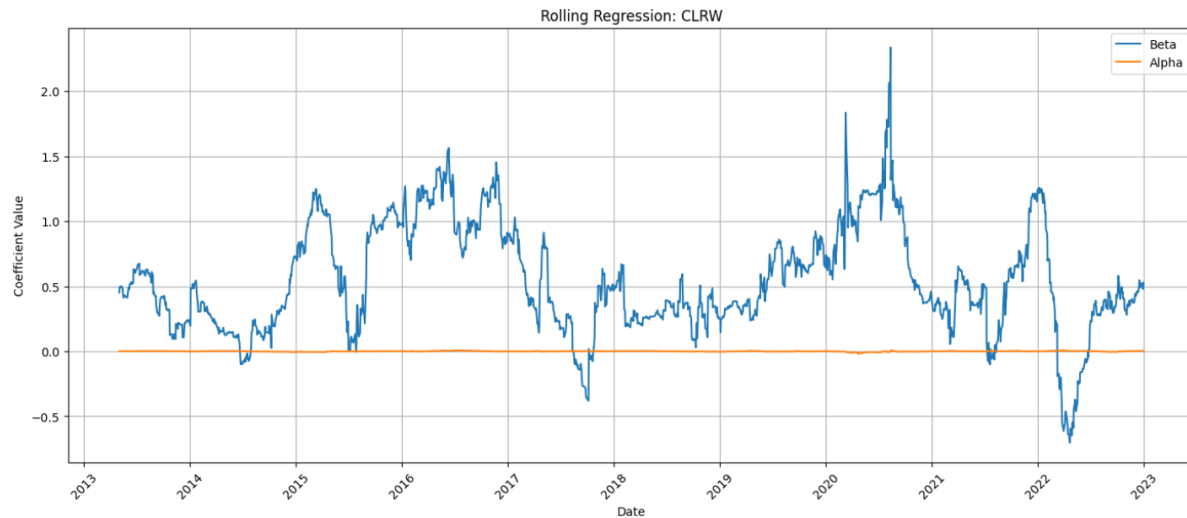


Figure 4: Rolling Regression of CL Futures Plot

Gold's beta demonstrates significant volatility throughout the period, frequently oscillating between positive and negative values. This pattern reveals gold's inconsistent relationship with the S&P 500, sometimes functioning as a market hedge (negative beta) and other times moving with equities.

The ES futures displays high market correlation with beta always hovering near the 1 value, indicating strong alignment with the benchmark as expected.

Oil exhibits persistently positive beta with distinctive cyclical patterns, occasionally surging above 2.0 during periods of market stress. This reveals crude oil's heightened sensitivity to economic conditions.

4.3 Random Forest Result

The following graphs showcase the rolling regression plotted onto a graph, all completed within python:

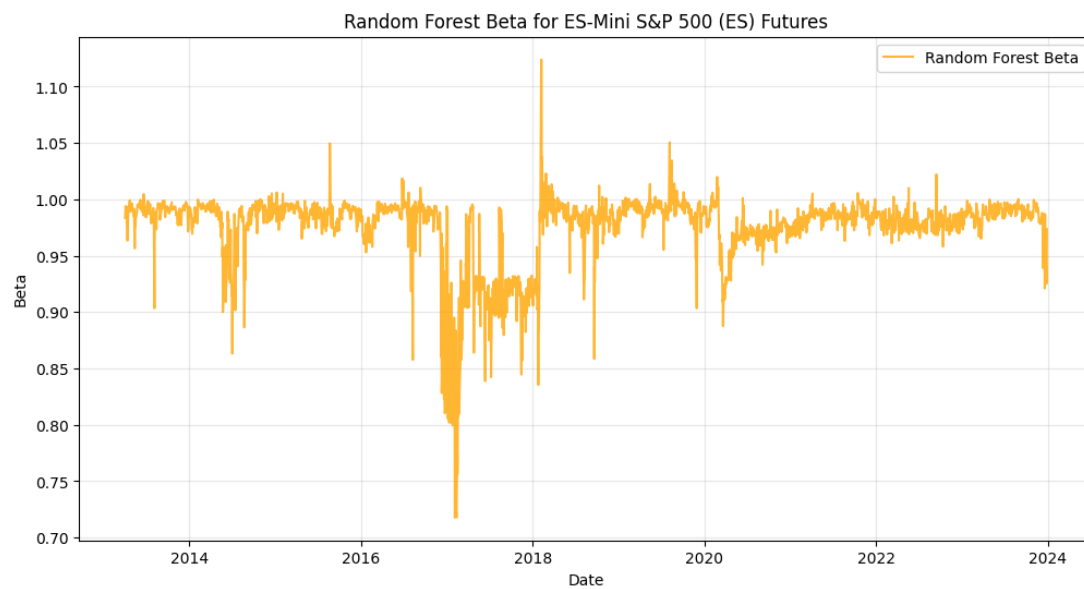


Figure 5: Random Forest Beta Graph Plot for ES Futures

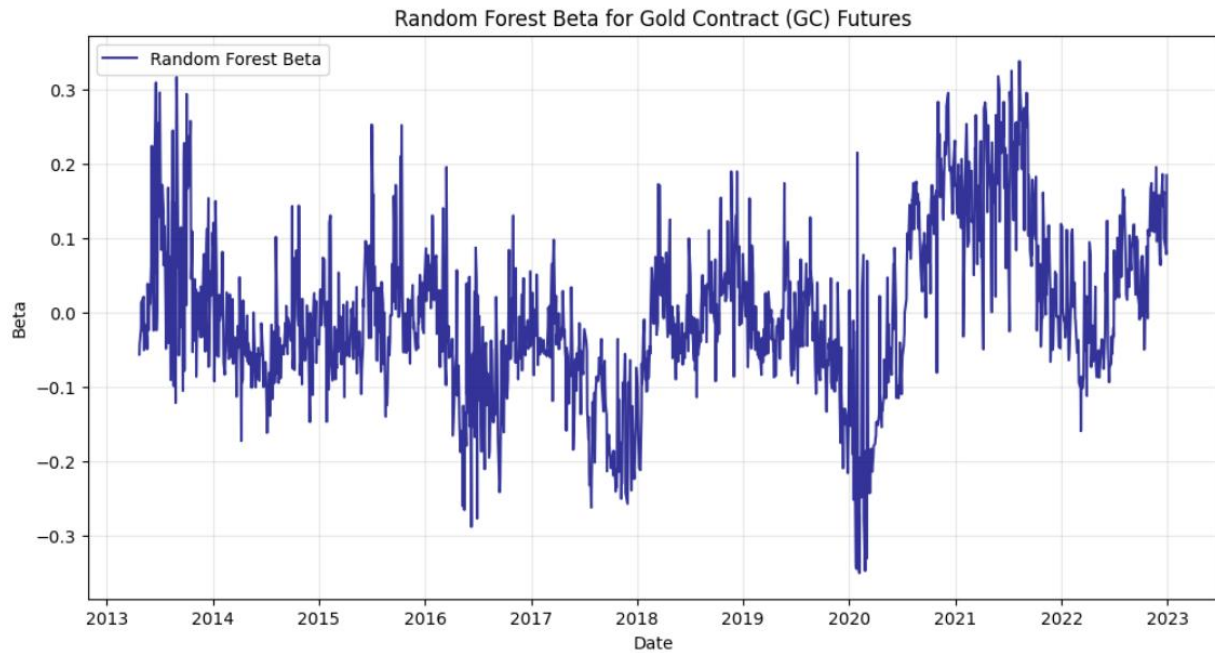


Figure 6: Random Forest Beta Graph Plot for GC Futures

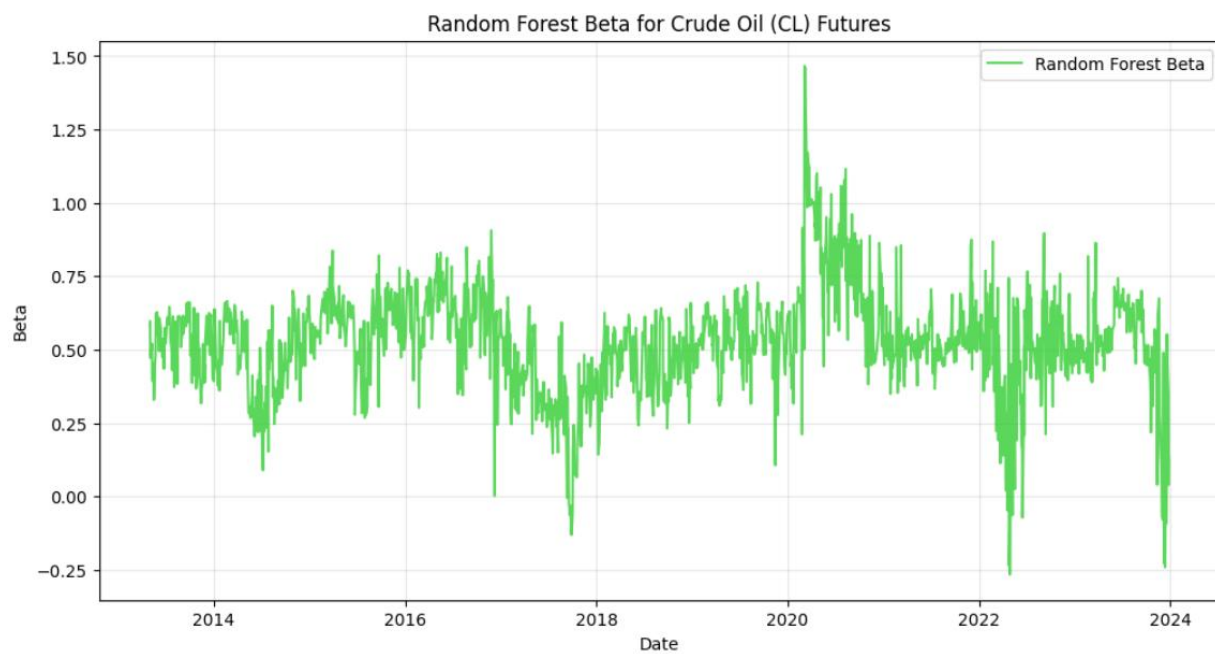


Figure 7: Random Forest Beta Graph Plot for CL Futures

For all three of the futures contracts, the Random Forest graphs had an extremely similar shape and structure in comparison to the Rolling Window Regression Graphs.

4.4 Back Testing Predicted vs. Actual Returns

As aforementioned, the CAPM formula will be used to calculate the predicted returns. This will then be compared to the actual returns to derive which method was most successful.

The results from this back testing process are as such:

Futures Contracts	Beta-Estimation Model	Mean Squared Error (MSE)	Mean Absolute Error (MAE)
ES	Fixed-Time Period	0.000008	0.001122
	Rolling Windows	0.000007	0.001119
	Random Forest	0.000008	0.001149
GC	Fixed-Time Period	0.000073	0.005889
	Rolling Windows	0.000068	0.005815
	Random Forest	0.00007	0.00584
CL	Fixed-Time Period	0.000611	0.015036
	Rolling Windows	0.000579	0.014785
	Random Forest	0.000593	0.014913

Figure 8: Table of MSE & MAE Values Across all Beta Estimation Models

Across all futures contracts examined, the Rolling Window method consistently demonstrated superior performance for all three futures contracts, yielding the lowest Mean Squared Error (MSE) and Mean Absolute Error (MAE) as shown in the above table.

Random Forest produced smoother beta trajectories while still capturing significant volatility-driven spikes. However, this method still slightly underperformed Rolling Window, likely due to noisy target beta values and limited inputs. The Fixed Time Period

approach failed to adapt to changing market conditions, resulting in the highest prediction errors across all contracts.

5 Discussion

The results support my initial hypothesis as the rolling window method's superior adaptability makes it particularly well-suited for dynamic risk estimation within financial markets. While slightly less effective, Random Forest shows promising potential for enhancement through additional predictive features such as futures-specific volatility measures. The Fixed Time Period approach proved least effective, highlighting the importance of accounting for time-varying risk within the markets. For the fund, this can have huge implications and can alter the way we approach beta estimation.

6 Conclusion

This research emphasizes the importance of dynamic capabilities in beta estimation for futures contracts. For a potential future extension, I would definitely explore implementing more inputs to the Random Forest model as more inputs could yield a more accurate model. Furthermore, the consistently higher errors observed for CL suggest that the S&P 500 may serve as a suboptimal benchmark for oil futures, and for future research, potentially altering the benchmarks could result in less variability.

7 References

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The code used can be found here: <https://github.com/lucasjohn05/BetaEstimator>

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